

Experiment 1

Voltage Regulation using Zener Diode and IC 7805

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April 19, 2026

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1 Aim

Study of Zener diode as a voltage regulator and use of IC 7805 voltage stabilizer.

2 Theory

Voltage regulation ensures that a circuit provides a constant output voltage regardless of changes in input voltage or load conditions.

2.1 Zener Diode

A Zener diode is a specialized type of diode designed to reliably allow current to flow "backwards" when a certain set reverse voltage, known as the Zener voltage (V_Z), is reached.

- **Reverse Bias Breakdown:** In forward bias, it behaves like a standard diode. In reverse bias, it conducts very little current until the breakdown voltage is reached. At this point (V_Z), the diode undergoes Zener breakdown (or Avalanche breakdown for higher voltages), causing the current to increase rapidly while the voltage across it remains nearly constant.
- **Voltage Regulation:** Because the voltage drop across the diode remains stable at V_Z over a wide range of reverse currents, it can be placed in parallel with a load to maintain a fixed voltage, protecting the load from voltage fluctuations.

2.2 IC 7805

The LM7805 is a popular 3-terminal positive voltage regulator IC from the 78xx series.

- **Function:** It accepts an unregulated DC input voltage (typically $> 7V$) and provides a stable, fixed +5V DC output.
- **Pins:**
 1. **Input (Pin 1):** Receives the unregulated input voltage.
 2. **Ground (Pin 2):** Connected to the common ground.
 3. **Output (Pin 3):** Delivers the regulated +5V output.
- **Advantages:** Unlike a simple Zener regulator, the 7805 includes internal current limiting, thermal shutdown, and safe-area protection, making it virtually immune to damage from output overloads.

2.3 Line Regulation

The ability to maintain a constant output voltage V_o when the input voltage V_i varies while the load is constant is called line regulation. Ideally, the change in output voltage (ΔV_o) should be zero for a change in input voltage (ΔV_i).

2.4 Load Regulation

The ability to maintain a constant output voltage V_o when the load resistance R_L varies while the input voltage V_i is constant is called load regulation. As the load current (I_L) increases, a good regulator will keep V_o constant.

3 Experimental Setup

The experiment was conducted using a DC power supply, a breadboard, and digital multimeters.

3.1 Zener Diode Regulator Circuit

A Zener diode of breakdown voltage V_Z is reverse connected to an input voltage source V_i across a load resistance R_L and a series resistor R_S . The voltage across the Zener will remain steady at its breakdown voltage V_Z for all values of Zener current i_Z as long as the current remains in the breakdown region.

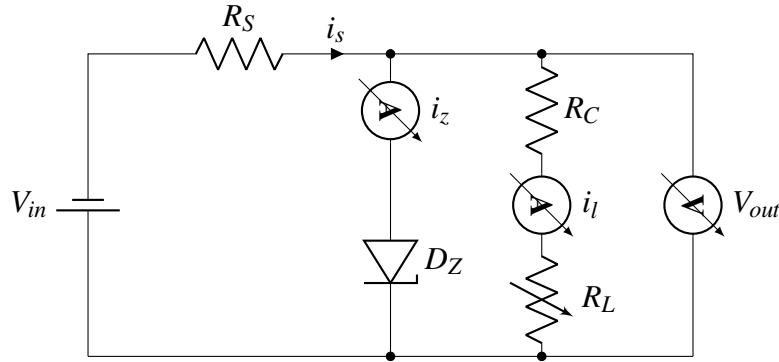


Figure 1: Circuit Diagram for Zener Diode Voltage Regulator

3.2 IC 7805 Regulator Circuit

The IC 7805 is a specialized three-pin integrated circuit designed for fixed 5V regulation. Pin 1 is for Input, Pin 2 is Ground, and Pin 3 provides the regulated Output.

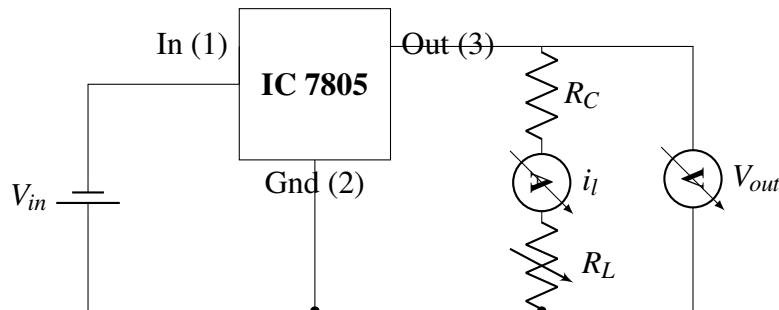


Figure 2: Circuit Diagram for IC 7805 Voltage Regulator

4 Data

4.1 Part A: Zener Diode Regulator

4.1.1 A-I: Line Regulation

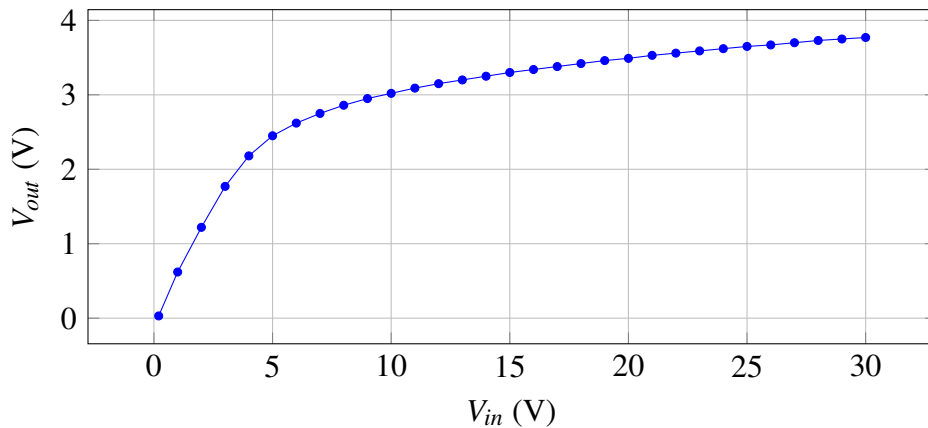
Condition: In line regulation, series resistance R_S and load resistance R_L are fixed, only input voltage V_{in} is varied. Any change in V_{in} appears across the series limiting resistance R_S .

$$\delta i_Z = \delta i_S$$

Table 1: Zener Line Regulation Data

V_{in} (V)	i_S (mA)	i_Z (mA)	V_o (V)	V_{in} (V)	i_S (mA)	i_Z (mA)	V_o (V)
0.2	0.00	0.00	0.03	16.0	5.00	4.85	3.34
1.0	0.00	0.00	0.62	17.0	5.43	5.28	3.38
2.0	0.00	0.00	1.22	18.0	5.90	5.73	3.42
3.0	0.03	0.03	1.77	19.0	6.34	6.16	3.46
4.0	0.16	0.17	2.18	20.0	6.81	6.62	3.49
5.0	0.43	0.43	2.45	21.0	7.29	7.07	3.53
6.0	0.77	0.76	2.62	22.0	7.76	7.54	3.56
7.0	1.15	1.12	2.75	23.0	8.23	7.99	3.59
8.0	1.54	1.50	2.86	24.0	8.68	8.44	3.62
9.0	1.96	1.89	2.95	25.0	9.16	8.91	3.65
10.0	2.37	2.30	3.02	26.0	9.64	9.37	3.67
11.0	2.80	2.71	3.09	27.0	10.11	9.84	3.70
12.0	3.22	3.13	3.15	28.0	10.61	10.30	3.73
13.0	3.66	3.56	3.20	29.0	11.10	10.78	3.75
14.0	4.10	3.98	3.25	30.0	11.58	11.24	3.77
15.0	4.55	4.42	3.30				

AI: Zener Line Regulation



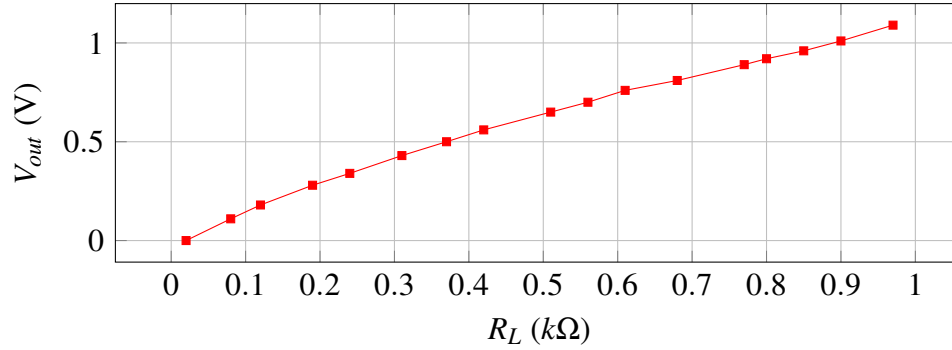
4.1.2 A-II: Load Regulation (Without Series Limiting Resistor R_C)

Condition: Since the voltage V_Z across the Zener remains constant, i_S is independent of load. Hence in this case:

$$\delta i_Z = -\delta i_L$$

Table 2: Without R_C

V_i (V)	R_L ($k\Omega$)	i_L (mA)	i_Z (mA)	V_o (V)
15.02	0.02	1.50	3.92	0.00
15.02	0.08	1.46	3.97	0.11
15.02	0.12	1.43	4.00	0.18
15.02	0.19	1.40	4.04	0.28
15.02	0.24	1.35	4.06	0.34
15.02	0.31	1.32	4.10	0.43
15.02	0.37	1.30	4.12	0.50
15.02	0.42	1.26	4.15	0.56
15.02	0.51	1.20	4.19	0.65
15.02	0.56	1.16	4.21	0.70
15.02	0.61	1.14	4.23	0.76
15.02	0.68	1.11	4.26	0.81
15.02	0.77	1.10	4.29	0.89
15.02	0.80	1.07	4.30	0.92
15.02	0.85	1.06	4.31	0.96
15.02	0.90	1.04	4.34	1.01
15.02	0.97	1.00	4.36	1.09

A-II: Load Regulation (No R_C)

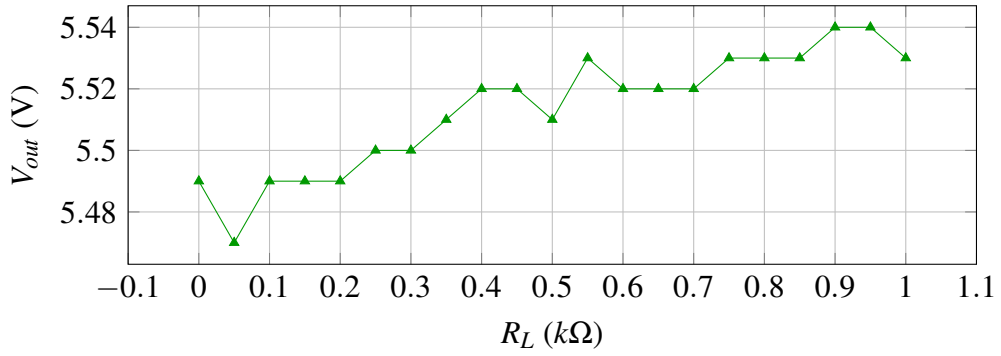
4.1.3 A-III: Load Regulation (With Series Limiting Resistor R_C)

Condition: Using a resistor $R_C = 2.2k\Omega$.

Table 3: With $R_C = 2.2k\Omega$

V_i (V)	R_L ($k\Omega$)	I_l (mA)	I_z (mA)	V_o (V)
15	0.00	2.52	1.91	5.49
15	0.05	2.47	1.95	5.47
15	0.10	2.41	2.01	5.49
15	0.15	2.36	2.05	5.49
15	0.20	2.31	2.12	5.49
15	0.25	2.26	2.17	5.50
15	0.30	2.23	2.21	5.50
15	0.35	2.17	2.27	5.51
15	0.40	2.14	2.29	5.52
15	0.45	2.10	2.32	5.52
15	0.50	2.06	2.38	5.51
15	0.55	2.00	2.41	5.53
15	0.60	1.99	2.45	5.52
15	0.65	1.96	2.47	5.52
15	0.70	1.92	2.51	5.52
15	0.75	1.90	2.52	5.53
15	0.80	1.86	2.57	5.53
15	0.85	1.83	2.58	5.53
15	0.90	1.81	2.62	5.54
15	0.95	1.79	2.64	5.54
15	1.00	1.75	2.68	5.53

A-III: Load Regulation (With R_C)



4.2 Part B: IC 7805 Voltage Regulator

4.2.1 B-I: Line Regulation

Condition: Constant load resistance.

Table 4: IC 7805 Line Regulation

$R_C + R_L$ (k Ω)	V_{in} (V)	i_L (mA)	V_{out} (V)
2.2	0.19	0.38	0.01
2.2	1.00	0.20	0.01
2.2	2.00	0.75	1.27
2.2	3.00	1.13	2.18
2.2	4.00	1.56	3.09
2.2	5.00	1.96	4.01
2.2	6.00	2.37	4.92
2.2	7.00	2.41	5.02
2.2	8.00	2.40	5.02
2.2	9.00	2.40	5.02
2.2	10.00	2.40	5.02
2.2	11.00	2.40	5.02
2.2	12.00	2.40	5.02
2.2	13.00	2.40	5.02
2.2	14.00	2.40	5.02
2.2	15.00	2.40	5.02
2.2	16.00	2.40	5.02
2.2	17.00	2.40	5.02
2.2	18.00	2.39	5.02
2.2	19.00	2.40	5.02
2.2	20.00	2.40	5.02

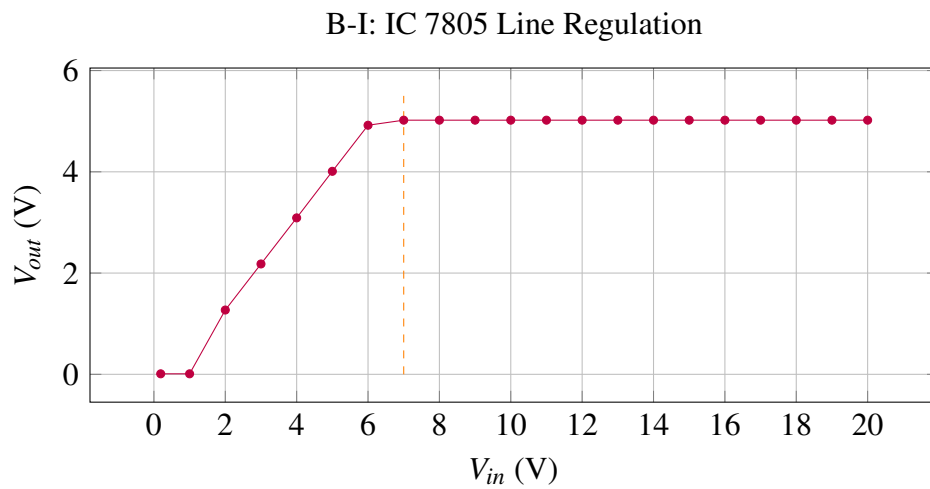


Figure 3: Output stabilizes at 5.02V for $V_{in} \geq 7V$.

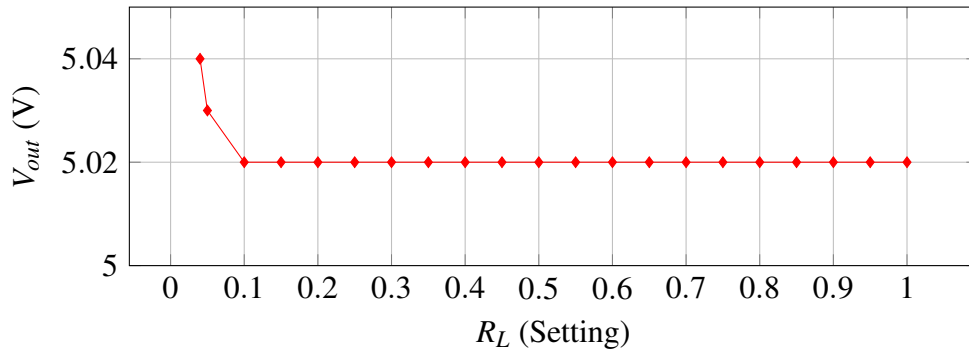
4.2.2 B-II: Load Regulation

Condition: Constant Input Voltage $V_{in} = 15.0\text{V}$.

Table 5: IC 7805 Load Regulation

V_{in}	R_L	i_L	V_{out}	V_{in}	R_L	i_L	V_{out}
15.0	0.04	2.36	5.04	15.0	0.55	1.89	5.02
15.0	0.05	2.27	5.03	15.0	0.60	1.85	5.02
15.0	0.10	2.22	5.02	15.0	0.65	1.82	5.02
15.0	0.15	2.17	5.02	15.0	0.70	1.79	5.02
15.0	0.20	2.12	5.02	15.0	0.75	1.82	5.02
15.0	0.25	2.08	5.02	15.0	0.80	1.77	5.02
15.0	0.30	2.04	5.02	15.0	0.85	1.73	5.02
15.0	0.35	2.00	5.02	15.0	0.90	1.69	5.02
15.0	0.40	1.99	5.02	15.0	0.95	1.66	5.02
15.0	0.45	1.96	5.02	15.0	1.00	1.63	5.02
15.0	0.50	1.93	5.02				

B-II: IC 7805 Load Regulation



5 Results

In the experiment we found:

1. In case of line regulation using Zener diode, $\delta i_Z = \delta i_S$.
2. Breakdown voltage of the Zener diode used in line regulation, $V_Z = 2.85\text{ V}$.
3. In case of load regulation using Zener diode, $\delta i_Z = -\delta i_L$.
4. Breakdown voltage of the Zener diode used in load regulation, $V_Z = 5.48\text{ V}$.
5. IC 7805 is much more better as a voltage regulator than a Zener diode.

6 Conclusion

This experiment demonstrated how voltage regulation works using a Zener diode and an IC 7805. We saw that the Zener diode can keep the voltage fairly constant once it hits its breakdown region, but it struggles if the load current gets too high (unless a series resistor like R_C is added to protect it).

In comparison, the IC 7805 is much more stable. It kept the output locked at 5.02 V very precisely, as long as the input voltage was at least 7V. This shows why dedicated ICs are preferred for reliable power supplies in electronics.